

Reviewer Pack — Minimal Rank of Quantum Group Representations and Communicat...

Entry 1a49e4686e18 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quantum Group Representations and Communication Complexity Rank

Entry ID: 1a49e4686e18 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-03 07:51:54 UTC

1. Conjecture

Field A (mathematical branch): Quantum Group Theory **Field B** (complexity object): Communication Complexity

Statement:

For every d -regular graph G , the minimal rank of the quantum group representation $R(G)$ associated with its vertex set is linearly correlated with its communication complexity rank $w(G)$, such that $\text{Rank}(R(G)) = \Theta(w(G))$. Furthermore, there exists a constructive mapping from each instance of G to a quantum group representation $R(G)$ such that $|\text{Rank}(R(G)) - w(G)| \leq k$.

Rationale (proposer's reasoning):

Quantum groups are noncommutative generalizations of Lie algebras and have been studied in the context of qubits and quantum computing. Their representations can be used to encode complex systems, which may provide a bridge to communication complexity. A strong correlation between these ranks could imply that certain communication tasks require high-dimensional encoding, suggesting a potential hardness of approximation.

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): c4a219a3ad76b4be

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the correlation coefficient from linear regression between $\text{Rank}(R(G))$ and $w(G)$ for all generated d -regular graphs G is greater than or equal to 0.8 AND the mean absolute difference $|\text{Rank}(R(G)) - w(G)|$ across all seeds is less than or equal to k .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.70	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def generate_d_regular_graph(n, d):
    if (n * d) % 2 != 0:
        return None
    graph = {i: [] for i in range(n)}
    edges = set()
    for _ in range(d * n // 2):
        u = random.randint(0, n - 1)
        v = random.randint(0, n - 1)
        if u == v or (u, v) in edges or (v, u) in edges:
            continue
        graph[u].append(v)
        graph[v].append(u)
        edges.add((u, v))
    return graph

def communication_complexity_rank(graph):
    n = len(graph)
    rank = 0
    for i in range(n):
        neighbors = set(graph[i])
        if not neighbors:
            continue
        min_degree = min(len(graph[j]) for j in neighbors)
        rank += min_degree
    return rank

def quantum_group_representation_rank(graph):
    n = len(graph)
    rank = 0
    for i in range(n):
        neighbors = set(graph[i])
        if not neighbors:
            continue
```

```

        rank += len(neighbors)
    return rank

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        graph = generate_d_regular_graph(n, 3)
        if graph is None:
            continue
        rank_qg = quantum_group_representation_rank(graph)
        rank_comm = communication_complexity_rank(graph)
        results.append((rank_qg, rank_comm))

    if not results:
        return {
            "metric_name": "Rank Difference",
            "metric_value": 0,
            "instances_tested": 0,
            "n_max": 0,
            "conjecture_holds": False,
            "counterexample": "mapping_undefined"
        }

    mean_diff = sum(abs(qg - comm) for qg, comm in results) / len(results)
    return {
        "metric_name": "Rank Difference",
        "metric_value": mean_diff,
        "instances_tested": len(results),
        "n_max": max(n_values),
        "conjecture_holds": mean_diff <= 1,
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_diff = sum(result["metric_value"] for result in results) / len(results)
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_diff} std=0 support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_diff} std=0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
metric_name': 'Rank Difference', 'metric_value': 5.75, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 8.25, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 8.25, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 5.25, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 6.0, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 6.5, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 7.75, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 8.5, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 11.25, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 7.25, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 3.25, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 5.25, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
TRIAL: {'metric_name': 'Rank Difference', 'metric_value': 8.5, 'instances_tested': 4, 'n_max': 40, 'conjecture': 'The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.'}
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test only considers a small range of n (up to 40) and the metrics used are not well-defined for the purpose of the conjecture. The quantum group representation rank is computed as the sum of the degrees of vertices, which does not correspond to the minimal rank of the quantum group representation. The communication complexity rank is also computed in a way that does not align with its standard definition.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that the conjecture does not hold for at least one seed (first_failing_seed=11) with a counterexample 'mapping_undefined'. A | next: Reconsider the definition of the quantum group representation rank and communication complexity rank to ensure they align with standard mathematical definitions. Expand the range of n tested and reevaluate the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 12

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	18821
2	propose	ollama_remote	glm4:latest	0	0	18938
3	propose	ollama_remote	glm4:latest	0	0	20854
4	preregistration	ollama_remote	glm4:latest	0	0	9316
5	novelty	ollama_remote	glm4:latest	0	0	16204
6	novelty	ollama_remote	glm4:latest	0	0	8511
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14879

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15138
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18439
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13071
11	critic	ollama_remote	glm4:latest	0	0	15759
12	judge	ollama_remote	glm4:latest	0	0	13860

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 183788 ms total latency. Provider mix: {'ollama_remote': 12}

(full prompt+response transcripts available in *research/audit/1a49e4686e18.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/1a49e4686e18.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/1a49e4686e18.tar.gz* (if generated)